

Advanced (Habanero)

- Q1.** What is the greatest common factor of 12 and 18?
(A) 2
(B) 3
(C) 6
(D) 9
(E) 12
- Q2.** Factor out the greatest common factor from $8x + 12y$
(A) $2(4x + 6y)$
(B) $4(2x + 3y)$
(C) $8(x + y)$
(D) $12(x + y)$
(E) $4x + 6y$
- Q3.** What is the factored form of $9x^2 + 12x$?
(A) $3x(3x + 4)$
(B) $3x(3x - 4)$
(C) $3(3x^2 + 4x)$
(D) $3(3x^2 - 4x)$
(E) $9x^2 + 12x$
- Q4.** What is the greatest common factor of 20 and 30?
(A) 2
(B) 5
(C) 10
(D) 15
(E) 20
- Q5.** Factor out the greatest common factor from $15a - 25b$
(A) $5(3a - 5b)$
(B) $5(3a + 5b)$
(C) $15(a - b)$
(D) $15(a + b)$
(E) $5a - 25b$
- Q6.** Simplify $6(2x + 3)$ using factoring.
(A) $12x + 9$
(B) $12x - 9$
(C) $12x + 18$
(D) $3(2x + 6)$
(E) $3(2x - 6)$
- Q7.** What is the factored form of $48x^3 + 18x^2$?
(A) $6x^2(8x + 3)$
(B) $6x(8x^2 + 3)$
(C) $6x^2(8x - 3)$
(D) $6x(8x^2 - 3)$
(E) $6x^2(3x + 8)$
- Q8.** Factor out the greatest common factor from $4x^2y + 12xy^2$.
(A) $4xy(x + 3y)$
(B) $4xy(x - 3y)$
(C) $12xy(x + y)$
(D) $12xy(x - y)$
(E) $4x^2y + 12xy^2$

Open Ended Questions (FRQ)

- Q9.** Factor out the greatest common factor of $12x^2y - 20xy^2 + 8xy$ and simplify your answer.
- Q10.** Factor the expression $24x^2y + 16x^2z + 40xy^2$ by finding the greatest common factor. Show your work and Simplify.

Chef's Tips

- Review all answers carefully for accuracy.
- Provide clear work and explanations where needed.
- No calculators or electronic devices allowed.